

Mobile Manipulation Platform for Imitation Learning

Zhuoyang Chen, Yuandi Huang

ABSTRACT:

Problems

- Rapid growth in learning-based robot control algorithms
- Lack of integrated platforms combining hardware, software and simulation
- Difficulties in collecting data for robot learning algorithms

Contributions

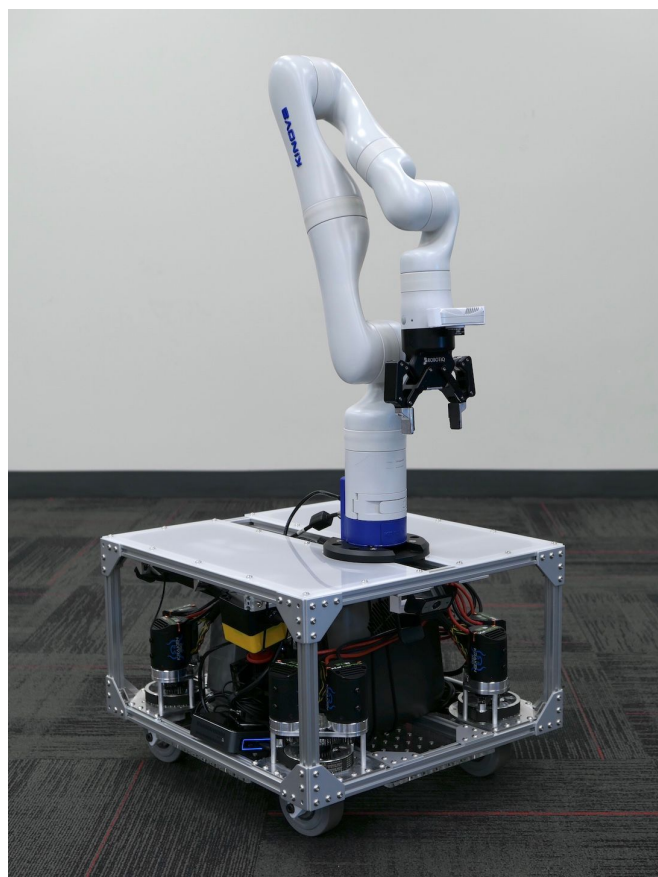
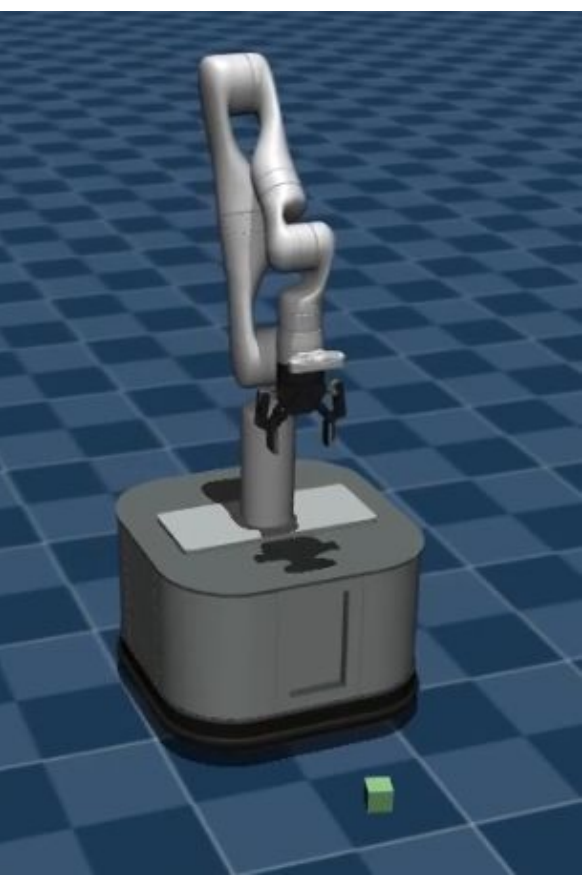
- Extended Tidybot++ mobile base manipulation platform
- Developed ROS2-based software stack with simulation, teleoperation and data collection
 - Flexible interface for experimentation with learning models
 - Bridges research and deployment in physical environments

PRIOR WORK:

Platform

Tidybot++ [1]

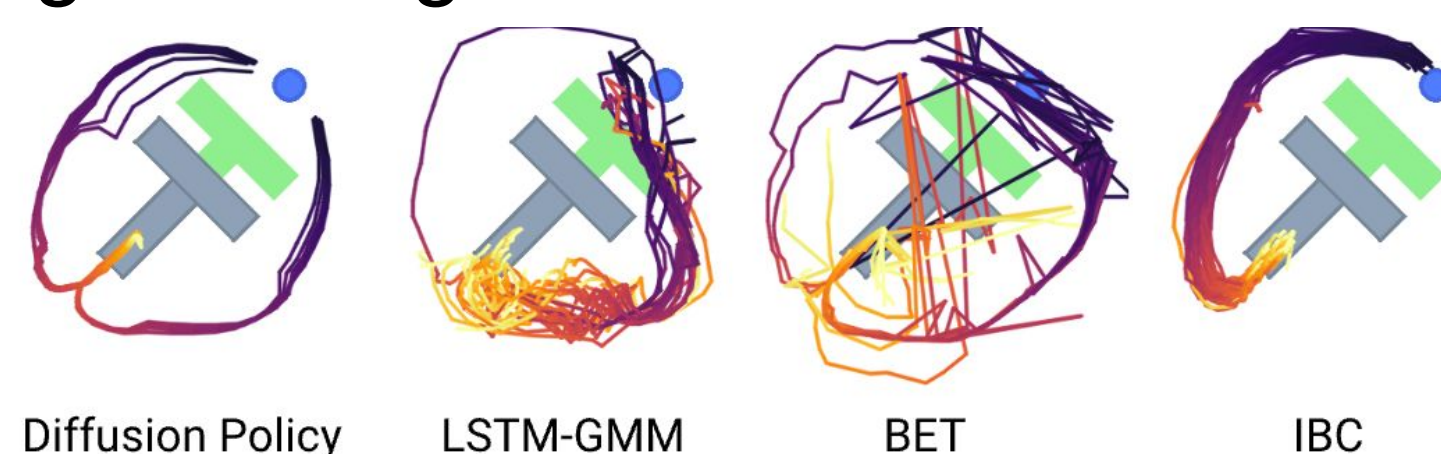
- Mobile manipulation robot
- Holonomic base with 3-DOF planar motion
- Phone teleoperation for training data collection



Algorithms

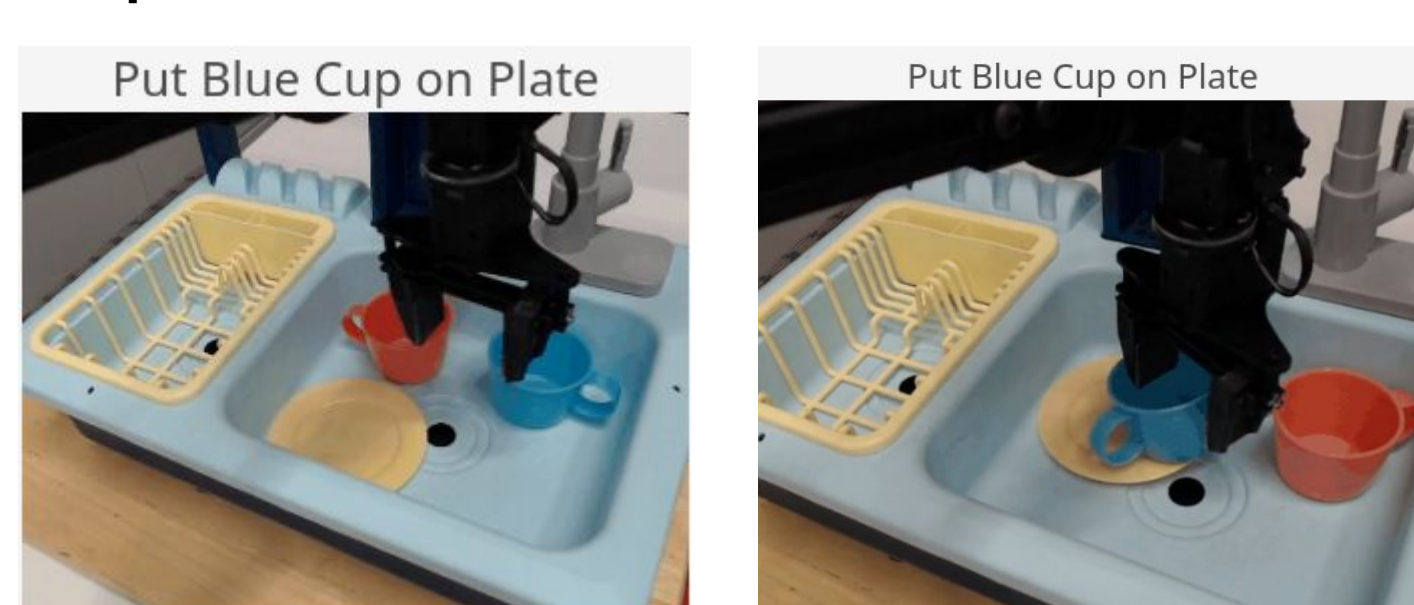
Diffusion Policy [2]:

Robot learning model capable of generating novel actions



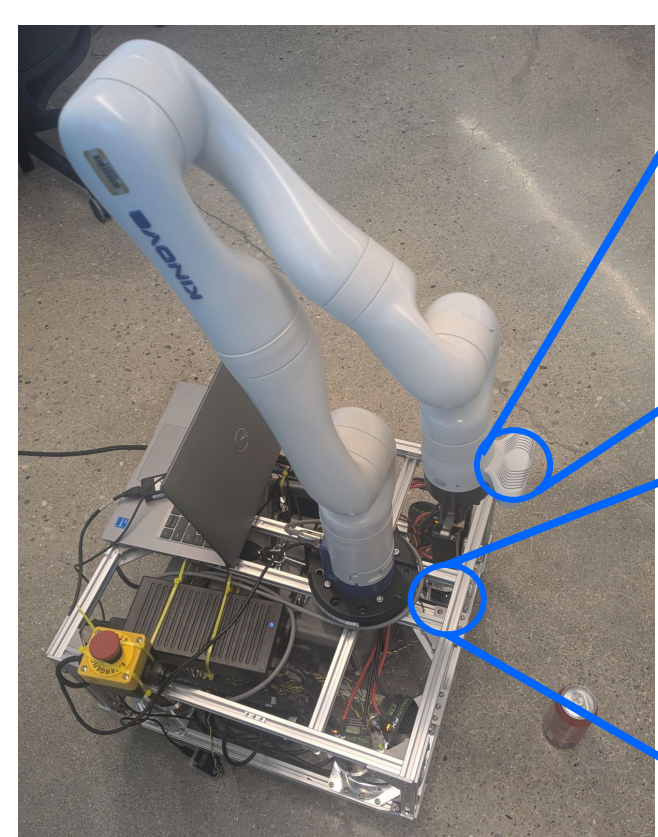
Vision-Language-Action [3]:

Large language model with vision capabilities



METHODS:

Teleoperate Robot



Phone is used for teleoperation

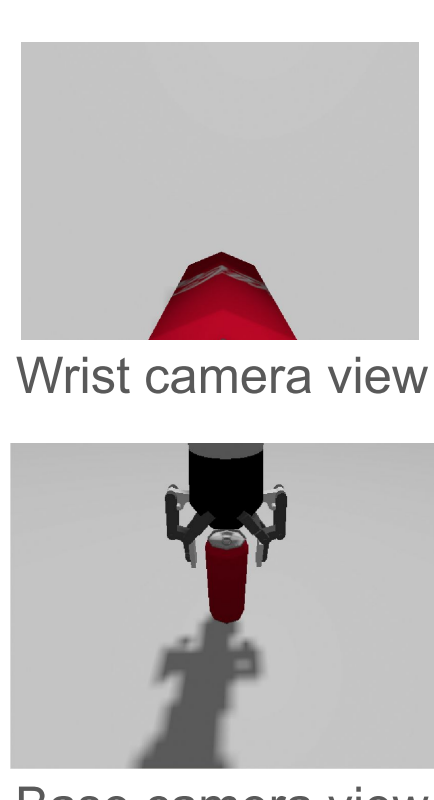
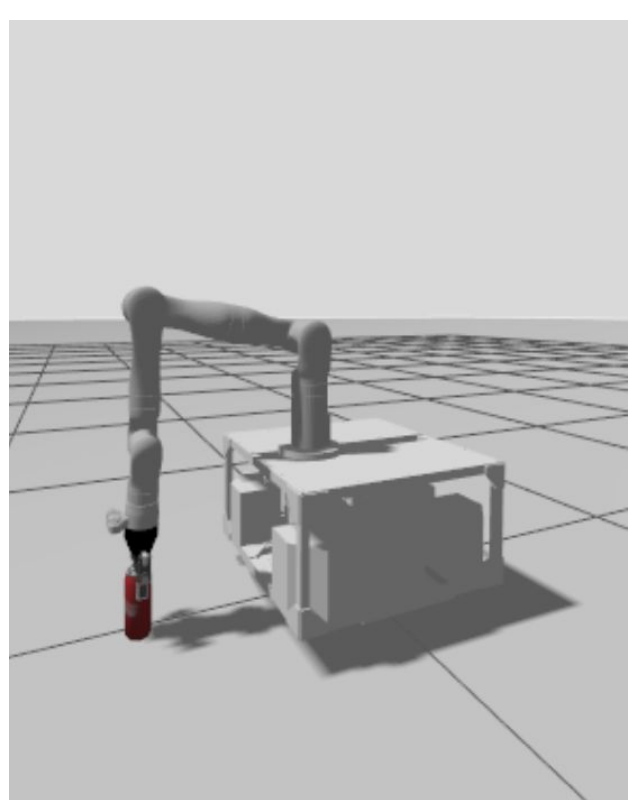
Validate Dataset



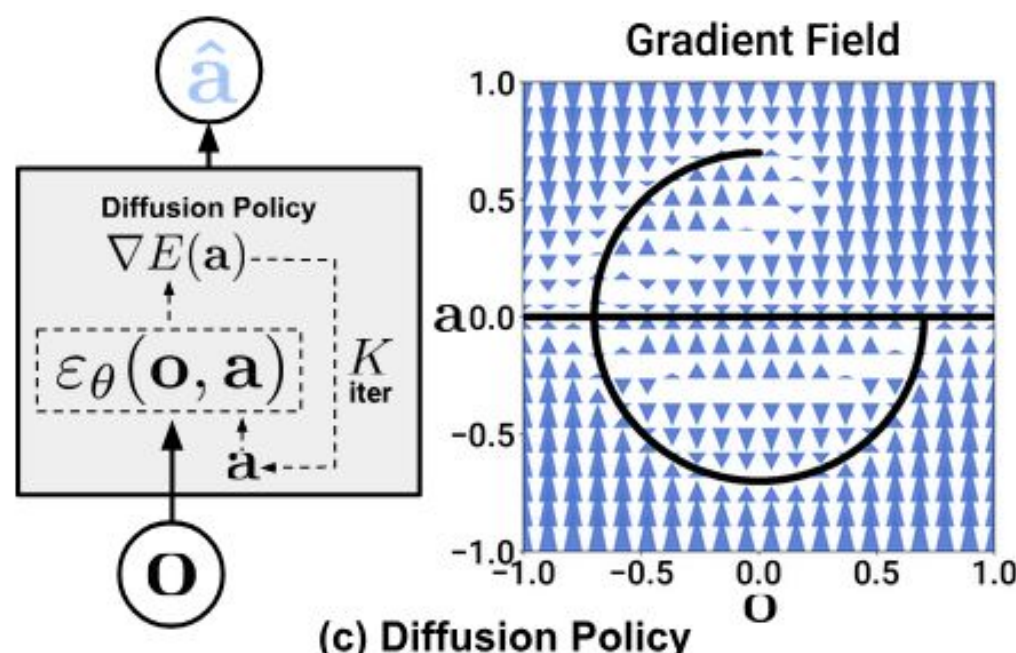
Commands and video feed are recorded



Verify Policy in Simulation



Train Diffusion Policy



Deploy to Physical Environment

PROJECT OBJECTIVES:

Mobile Manipulator Assembly

Construct a mobile manipulation platform

ROS2 Interface Development

A full ROS2 stack with the robot model, simulated environment, teleoperated control and hardware interface

Training Data Acquisition

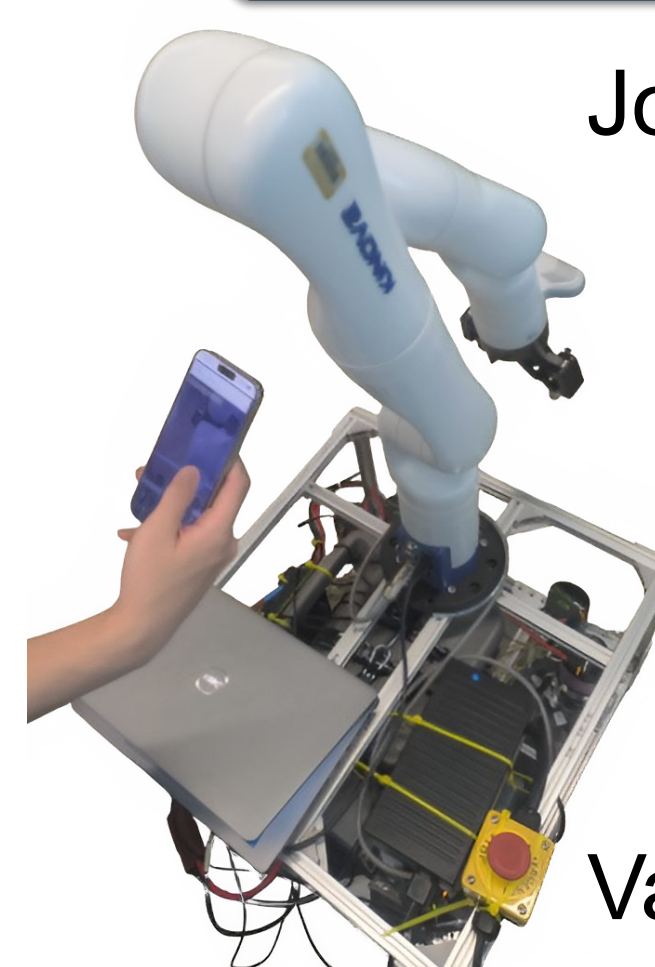
A framework to collect data from robot operation for use in training imitation learning models

PROJECT IMPACT:

- Kinova Gen3 Arm
- Robotiq-2f-85 gripper
- Powered caster wheel base
- RGBD wrist camera
- Onboard mini PC



Physical Hardware

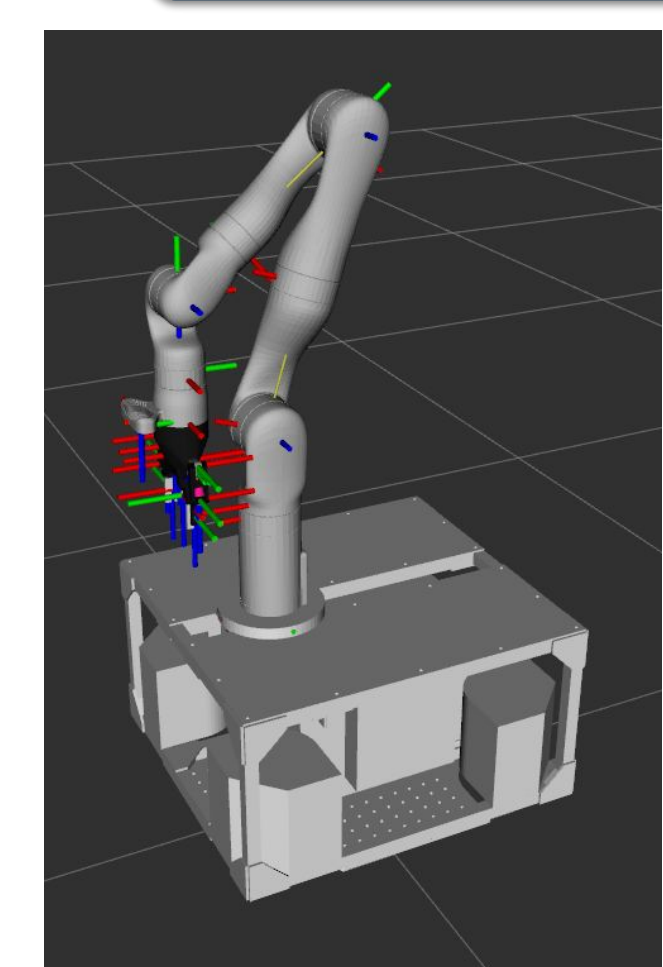


Joint Trajectories & Camera Feed

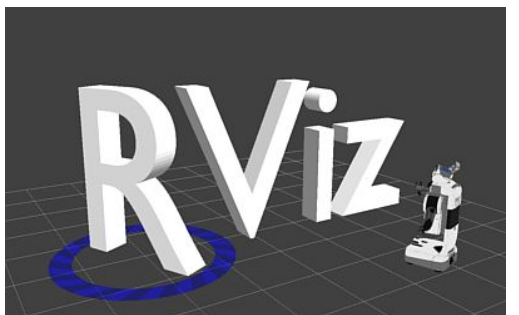
ROS bags

Valid dataset (.hdf5)

Simulated Environment



Movelt2



Training Data Acquisition

ROS2 Interface

Real-World Deployment of Imitation Learning Models:

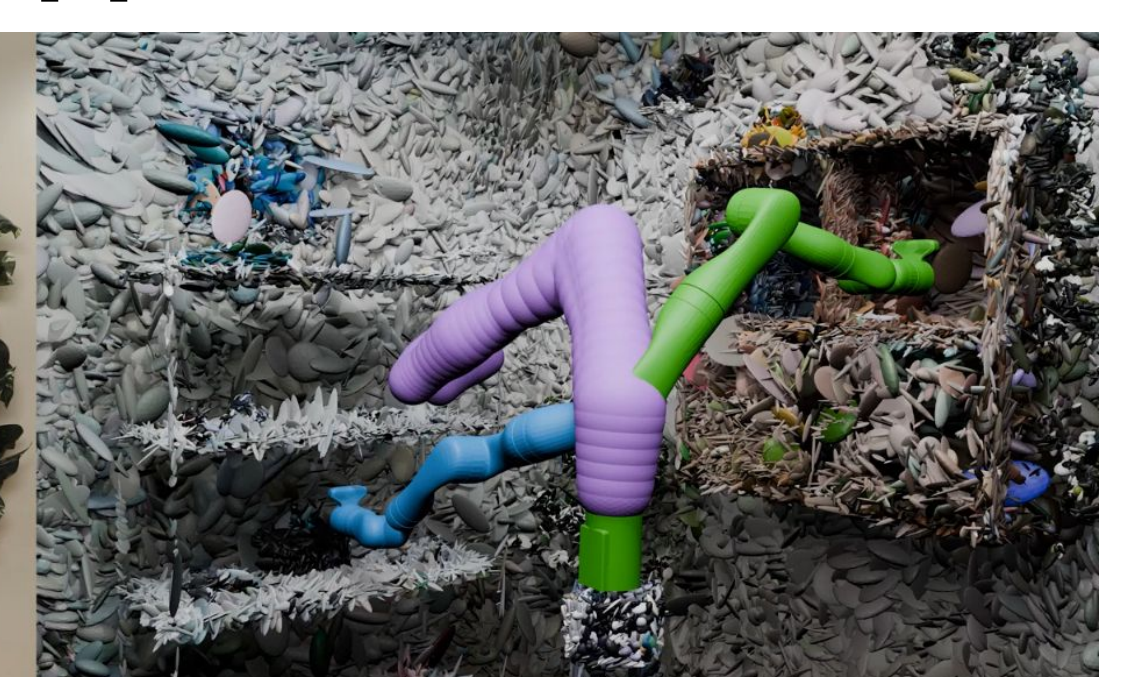
Enables systematic evaluation of how learned robot behaviors perform across diverse models

Platform for Learning Complex Behaviors:

Opens possibility for learning tasks requiring coordinated perception and manipulation, such as opening doors or natural language commands

FUTURE WORK:

- Collect data and train our own diffusion policy on the robot (open/close doors, pick-and-place)
- Equip the robot with LiDAR and SLAM systems for deployment of novel motion planning algorithms [4]



REFERENCES

- [1] J. Wu *et al.*, "TidyBot++: An Open-Source Holonomic Mobile Manipulator for Robot Learning," *arXiv preprint arXiv:2412.10447*, Dec. 2024, doi: 10.48550/arXiv.2412.10447.
- [2] C. Chi, Z. Xu *et al.*, "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion," *The International Journal of Robotics Research*, 2024.
- [3] M. J. Kim *et al.*, "OpenVLA: An Open-Source Vision-Language-Action Model," *arXiv preprint arXiv:2406.09246*, 2024.
- [4] J. Michaux *et al.*, "Let's Make a Splat: Risk-Aware Trajectory Optimization in a Normalized Gaussian Splat," *IEEE Transactions on Robotics*, 2025.

ACKNOWLEDGMENTS

We would like to thank our mentors, Seth Isaacson and Challen Enniful Adu, as well as Professor Ram Vasudevan for their guidance and support throughout this project.